

ASYMPTOTIC RS MCA

ABSTRACT. We give a conditional asymptotic Reed–Solomon mutual-correlated-agreement compiler from a first-match profile ledger. After quotient, planted, and other algebraic profiles have been routed, Balog–Szemerédi–Gowers and quasicube growth eliminate the high-energy branch of the primitive prefix problem, provided its Sidon-heavy branch is paid separately. We construct smooth rows with $\log |\mathbb{B}_n| = O(\log n) = o(n)$ having an exponentially large Sidon-heavy square-quotient prefix fiber at the identity crossing and, after scalar extension, exponentially many distinct MCA-bad slopes. The construction persists after stabilizer deletion, while its periodic form transfers to standard circle codes. Consequently the frontier theorem must use the full profile envelope, a Sidon payment on the routed primitive residual, and a distinct-ray compiler; the identity formula is recovered only under explicit identity-dominance and target-adjusted crossing hypotheses.

1. STATEMENT

Let $D \subseteq \mathbb{F}$ be an evaluation set of size n , and let

$$C = \text{RS}_{\mathbb{F}}(D, k) = \{(f(t))_{t \in D} : \deg f < k\}.$$

For $a \in \{0, \dots, n\}$, a finite slope $\gamma \in \mathbb{F}$ is support-wise MCA-bad for a pair $r_1, r_2 \in \mathbb{F}^D$ at agreement a if some set $S \subseteq D$, $|S| \geq a$, and some $h \in \mathbb{F}[X]$, $\deg h < k$, satisfy $r_1 + \gamma r_2 = h$ on S , while r_1, r_2 are not simultaneously explained on the same S by two degree- $< k$ polynomials. Put

$$B_C^{\text{MCA}}(a) = \max_{r_1, r_2} \#\{\gamma \in \mathbb{F} : \gamma \text{ is MCA-bad at agreement } a\}.$$

We use the full slope field, so $q_{\text{line}} = |\mathbb{F}|$, and set $\varepsilon_{\text{mca}}(C, 1 - a/n) = B_C^{\text{MCA}}(a)/|\mathbb{F}|$. Related finite-row normalizations appear in [Cho26a, Cho26b].

We consider smooth multiplicative or circle-type RS row sequences $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ with $|D_n| = n$, $\rho_n = k_n/n \rightarrow \rho \in (0, 1)$, and $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, where $\beta_n = \log_2 |\mathbb{B}_n| = o(n)$. For an agreement $a_n = k_n + 1 + w_n$, define the identity-prefix average

$$\overline{N}_{n, a_n} = \binom{n}{a_n} |\mathbb{B}_n|^{-w_n}.$$

For $r \in (0, 1)$ and $b \geq 0$, put

$$g^*(r, b) = \sup\{g \in [0, 1 - r] : H_2(r + g) \geq bg\},$$

where $H_2(x) = -x \log_2 x - (1 - x) \log_2 (1 - x)$, and write $g_n^* = g^*(\rho_n, \beta_n)$.

Here a multiplicatively smooth domain is a coset θH of a cyclic subgroup of $\mathbb{B}_n^\times$, together with its uniform-fiber power quotients. A circle-type domain is a Chebyshev image of a twin coset $gH \cup g^{-1}H$ in a norm-one torus, or a circle evaluation set obtained from such a multiplicative set by diagonal uniformization or stereographic parametrization. More generally, a polynomial map φ of degree d is a smooth quotient of D when every nonempty fiber $\varphi^{-1}(y) \cap D$ has exactly d elements. The quantitative assumptions used below are the stated profile-ledger and ray conditions; these geometric names specify the profile atlas to which they are applied.

For a fixed row and agreement, a *first-match profile atlas* is fixed before the received line is chosen. It is an ordered finite collection of constructible witness charts. A profile λ in the atlas specifies a full support slice Ω_λ^0 , a coefficient field $\mathbb{B}_\lambda$, and R_λ effective prefix equations; its natural average is $\overline{N}_\lambda = |\Omega_\lambda^0| |\mathbb{B}_\lambda|^{-R_\lambda}$. For a received line, $\Lambda(r_1, r_2; a)$ is the subcollection of fixed atlas profiles that occur under first match. Exhaustiveness of this atlas is an explicit ledger hypothesis, so profiles cannot be added or deleted after their sizes are known. The

identity profile is always included. Define the profile envelope

$$(1.1) \quad \mathfrak{E}_n(a) = 1 + (n - a + 1) + \sup_{r_1, r_2} \sum_{\lambda \in \Lambda(r_1, r_2; a)} \bar{N}_\lambda.$$

The tangent term is displayed because it cannot be absorbed when the identity average is below one, and $\mathfrak{E}_n(a) \geq 1 + (n - a + 1) + \bar{N}_{n,a}$. A window is *identity-dominant* if some $\kappa_n = \exp(o(n))$, uniformly in the window, satisfies

$$\mathfrak{E}_n(a) \leq \kappa_n(1 + (n - a + 1) + \bar{N}_{n,a}).$$

On the agreement lattice the target threshold is

$$\delta_C^*(\varepsilon) = \sup \left\{ 1 - \frac{a}{n} : a \in \{k + 1, \dots, n\}, \varepsilon_{\text{mca}}(C, 1 - a/n) \leq \varepsilon \right\},$$

whenever this set is nonempty.

Definition 1.1 (Cells and first-match payment). *A cell is a constructible subfamily of the MCA witness incidence space, or a bounded-complexity family of such subfamilies, stable under extension of the ground field. Cells are ordered. A slope is assigned to the first cell containing one of its witnesses. A nonprimitive algebraic cell is paid at agreement a_n if its distinct-slope image is at most its printed natural-profile budget; a Fourier/Sidon cell is paid in the support statistic of Definition 3.1. A profile family is paid if the sum of its first-match budgets is at most $\exp(o(n))\mathfrak{E}_n(a_n)$. Only in an identity-dominant window may this be specialized to the identity budget.*

Lemma 1.2 (First-match disjointization). *If ordered cells $\mathcal{C}_1, \dots, \mathcal{C}_J$ cover a set of bad slopes and the j -th first-match cell has budget U_j , then the covered slopes contribute at most $\sum_j U_j$ to $B_C^{\text{MCA}}(a)$.*

Proof. For a fixed received line, assign each bad slope to the least-indexed cell containing a witness. These assigned slope classes are disjoint, and the j -th class is contained in the projection of $\mathcal{C}_j$ after earlier cells have been removed. Summing the budgets and then maximizing over received lines proves the claim. $\square$

Definition 1.3 (Closed paid ledger). *A smooth or circle-type row sequence is closed-ledger at agreements a_n if the following first-match cells cover all MCA-bad witnesses except the primitive leaves of Definition 2.1, the number of profiles is $\exp(o(n))$, their natural coefficient fields and prefix depths are retained in (1.1), the algebraic cells have total distinct-slope contribution $\exp(o(n))\mathfrak{E}_n(a_n)$, and the Fourier/Sidon and primitive support statistics are paid at their declared profile scales.*

- (C1) Quotient-pullback cells. *The support or locator descends along a nontrivial finite map $D \rightarrow D'$. These cells include quotient-remainder locators, divisor-block support ledgers, quotient-image lcm ledgers, and quotient pullback recursion.*
- (C2) Chebyshev and dihedral cells. *On multiplicative or circle rows, the support is invariant under the dihedral action, or the locator factors through a Chebyshev or circle quotient such as $z + z^{-1}$.*
- (C3) Planted-block cells. *A positive-density block is forced into the support or common zero set. The payment is the entropy cost of planting that block, together with the bounded quotient profile.*
- (C4) Tangent and deep-center cells. *The received line lies in a tangent, secant, or high-contact stratum of the RS code variety, including the high-agreement tangent cell.*
- (C5) Extension and descent cells. *The witness either descends to a smaller field or is controlled by a bounded extension-line or extension-pole parameter.*
- (C6) Differential-locator and regular-rank cells. *The locator map has a rank drop, a ramification defect, or an overdetermined Hankel minor. These cells include regular closed-range Hankel packing, canonical rank-drop ledgers, and curve rank-drop ledgers.*

- (C7) Saturation and effective-image-collapse cells. *The apparent support count is larger than the actual line-ray or codeword-ray image. The payment uses exact saturation identities and exact residual image ledgers.*
- (C8) Balanced-core and split-pencil cells. *The residual witness lies in a balanced-core split-pencil chart. One-parameter primitive locator pencils are paid by the moving-root theorem; deficiency-one and identically split top charts are tangent-borne or eliminated. Every higher-dimensional residual chart requires a direct distinct-ray estimate.*
- (C9) Fourier/Sidon cells. *In a primitive prefix leaf, a heavy fiber with exponentially small additive closure is cut as a Fourier/Sidon cell: major arcs are routed to (C1)–(C8), and minor arcs are paid by Fourier flatness.*

The remaining first-match complement is a disjoint union of primitive prefix leaves. Every such leaf has the Boolean fixed-density Vandermonde form of Definition 2.1. If it is attached to profile λ , with residual mass M_λ , residual image size L_λ , and active length N_λ , closure requires

$$\log(M_\lambda/L_\lambda) = o(N_\lambda), \quad \frac{M_\lambda}{L_\lambda} \leq \exp(o(n))\bar{N}_\lambda.$$

The o -terms are uniform over the atlas. Its Sidon payment is imposed only after all earlier profiles have been removed. Thus a closed profile ledger includes the residual-to-full average comparison. Conversion of the resulting support budgets into distinct MCA slopes is the separate RC condition of Definition 1.4.

Definition 1.4 (Direct ray compiler). *A row satisfies RC at agreement a_n if the distinct slopes projected from every primitive residual profile are at most $\exp(o(n))$ times that profile's paid budget, with total cost $\exp(o(n))\mathfrak{E}_n(a_n)$. A max-fiber or support-pair estimate alone is not RC: it counts supports or pairs, not the image of the witness incidence in slope space.*

Theorem 1.5 (Cellwise ledger interface). *The results cited in [Cho26a, Cho26b] supply the proved cellwise constructions and estimates in (C1)–(C8) and the Fourier-flat part of (C9). For a given smooth or circle row they assemble to a closed profile ledger only when the atlas is exhaustive, every profile is charged at its natural scale in (1.1), and the residual-to-full average comparison holds, and the Sidon-heavy part of the routed primitive residual is paid. The full MCA compiler additionally requires RC. None of these cellwise results implies identity dominance.*

Proof. The cited cellwise estimates are as follows.

For (C1), the quotient-remainder lower and support ledgers, divisor-union support ledger, exact finite-parameter quotient-image ledger, affine/projective lcm ledgers, quotient pullback recursion, and bounded quotient census are proved in [Cho26a, quotient-remainder, quotient support, quotient-image, and bounded-census ledgers]. The corresponding first-match convention and quotient status cell are isolated in [Cho26b].

For (C2), exact Chebyshev fibers on circle twin-coset domains, circle quotient transport, and circle profile estimates are proved in [Cho26a, Chebyshev and circle sections]. Dihedral and Chebyshev cells are routed as quotient cells with extra monodromy; their payment still uses the natural-profile comparison required in the theorem.

For (C3), planted quotient profiles, planted quotient-core lower counts, planted list-budget windows, and bounded active quotient order are established in [Cho26a, planted quotient profile and bounded quotient census]. They identify the relevant planted entropy scale; the corresponding upper payment remains part of the closed-ledger hypothesis.

For (C4), the exact high-agreement tangent cell and tangent-borne split-kernel sections are proved in [Cho26a, high-agreement tangent exact cell and split-kernel tangent-borne theorem] and refined in [Cho26b, exact high-agreement tangent cell].

For (C5), the extension-pole deep-list and quotient-remainder floors, extension-line dimension-degree ledger, subfield confinement, and genuinely extension-valued line accounting are proved in [Cho26a, extension-pole and extension-line ledgers]; the full-extension cell target and extension-valued rank-one floor are recorded in [Cho26b].

For (C6), regular Hankel minors, canonical exact-agreement rank-drop ledgers, closed-ball rank-drop lcm ledgers, curve rank-drop ledgers, arbitrary-residual image ledgers, and the scanner residual ledger are proved in [Cho26a, Hankel, rank-drop, curve, residual-image, and scanner ledgers]. These are exactly the differential-locator and regular-rank payments.

For (C7), the exact saturation identity and line-ray saturation identity are proved in [Cho26b, saturation and line-ray saturation identities]; exact arbitrary-residual image and closed-ball residual image ledgers appear in [Cho26a, residual image ledgers]. Thus raw support overcounts are replaced by actual image counts.

For (C8), the interpolation-lattice split-pencil reduction, near-rational payment, deficiency-one split-test eliminant, split annihilator dictionary, tangent-borne split charts, boundary-profile identification with $\mathbb{Q}$, base-field split-pencil floor, one-parameter moving-root theorem, and primitive one-pencil corollary are proved in [Cho26b, split-pencil and BC sections]. The broader balanced-core and split-pencil reductions developed in [Cho26a, rank-one and split-pencil reductions] are used here as reductions to explicit split-pencil charts; higher-dimensional residual charts are still covered only when their distinct-ray estimates are supplied by the closed-ledger and RC hypotheses.

For (C9), Fourier-flat leaves satisfy asymptotic $\mathbb{Q}$ and large-characteristic Fourier examples are proved in [Cho26b, Fourier-flat $\mathbb{Q}$ section]. The Sidon cut and its high-energy alternative are stated and proved below. Fourier flatness pays the minor-arc branch after algebraic major arcs have been routed; it does not give a uniform identity-normalized payment before quotient, subfield, planted, and remainder profiles are removed. Recent C9 support packets prove several restricted payments and route cuts: block-profile Plotkin bounds, near-norm-gate remainder packing, coherent-phase Fourier subblock payment, and exclusions for radial-shell and low-complexity cube-parametrized routes [Dan26]. These packets clarify the residual target but do not by themselves exhaust (C9) or replace the Sidon payment required in Definition 3.1.

Finally, first-match disjointization is Lemma 1.2. These results give the asserted cellwise interface. Exhaustion, comparison at the natural profile scales, residual-to-full average comparison, residual Sidon payment, and projection from support statistics to distinct slopes are precisely the additional conditions in the statement. $\square$

Thus the cited geometry identifies and estimates the declared cells but does not unconditionally close an identity-scale ledger. The argument below proves the analytic primitive implication and the target compiler once the printed profile, Sidon, and ray conditions have been verified.

Remark 1.6 (Audit status). A line-by-line audit of the profile-envelope draft reports no problem with the obstruction construction, the natural-profile payment scale, or the separation between support statistics and ray counting [Bua26a]. It leaves as explicit proof obligations the imported window-uniformity input and the residual higher-dimensional split-pencil charts not reduced to the paid one-parameter, deficiency-one, or tangent-borne cases. The deployed-row numerical spine has also been checked independently as arithmetic support for the displayed profile-envelope crossing [Leg26]. Separate finite threshold packets for the KoalaBear and Mersenne rows sharpen current finite walls, including the star3 point-count and signed-cancellation target, but they are not used as hypotheses in the asymptotic theorem [Bua26b].

Theorem 1.7 (Conditional profile-envelope frontier). *Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ be a smooth or circle-type row as above, let $0 \leq \varepsilon_n \leq 1$, put $T_n = \lfloor \varepsilon_n |\mathbb{F}_n| \rfloor$, and let $\kappa_n = \exp(o(n))$ dominate the compiler losses uniformly in the window under consideration. Define the first safe agreement*

$$a_n^* = \min\{a \in \{k_n + 1, \dots, n\} : B_{C_n}^{\text{MCA}}(a) \leq T_n\}, \quad \delta_n^* = 1 - a_n^*/n,$$

whenever the set is nonempty. Suppose agreements $a_{-,n} < a_{+,n}$ satisfy:

- (i) *a closed profile ledger and RC hold at $a_{+,n}$, and $\kappa_n \mathfrak{E}_n(a_{+,n}) \leq T_n$;*
- (ii) *a certified identity, quotient, planted, or remainder construction gives $U_n(a_{-,n})$ distinct MCA-bad slopes, where $U_n(a_{-,n}) > T_n$.*

Then

$$(1.2) \quad a_{-,n} < a_n^* \leq a_{+,n}, \quad 1 - \frac{a_{+,n}}{n} \leq \delta_n^* < 1 - \frac{a_{-,n}}{n}.$$

If the two agreements are $o(n)$ apart around $a_n = k_n + 1 + g_n n$, then $\delta_n^* = 1 - \rho_n - g_n + o(1)$. For the identity construction, when $|\mathbb{F}_n| > n$, one may take

$$U_n(a) = \left\lceil \frac{L_n(a)(|\mathbb{F}_n| - n)}{|\mathbb{F}_n| - n + k_n(L_n(a) - 1)} \right\rceil, \quad L_n(a) = \left\lceil \binom{n}{a} |\mathbb{B}_n|^{-(a - k_n - 1)} \right\rceil.$$

For the identity-dominant specialization put

$$\tau_n = \frac{1}{n} \log_2(1 + T_n), \quad F_n(g) = H_2(\rho_n + g) - \beta_n g,$$

and define the target-adjusted right crossing by

$$(1.3) \quad g_{T,n} = \sup(\{g \in [0, 1 - \rho_n] : F_n(g) \geq \tau_n\} \cup \{0\}).$$

At an interior crossing,

$$H_2(\rho_n + g_{T,n}) - \beta_n g_{T,n} = \frac{1}{n} \log_2(1 + T_n).$$

If identity dominance and the collision-aware lower construction supply agreements satisfying (i)–(ii) with $a_{\pm,n} = k_n + 1 + g_{T,n}n + o(n)$, then $\delta_n^* = 1 - \rho_n - g_{T,n} + o(1)$. If additionally $T_n = \exp(o(n))$, then $g_{T,n} = g^*(\rho_n, \beta_n) + o(1)$, and hence

$$(1.4) \quad \delta_n^* = 1 - \rho_n - g^*(\rho_n, \beta_n) + o(1).$$

The rest of the paper proves this conditional implication from the cellwise interface and the external additive-combinatorics facts stated in Section 4. The counterexample below shows that identity dominance and any pre-routing uniform Sidon payment cannot be assumed. Residual Sidon payment remains a separate hypothesis, while Remark 6.3 explains independently why RC is required.

2. PRIMITIVE LEAVES

Definition 2.1 (Primitive prefix leaf). *A primitive leaf consists of a finite set T , $|T| = N$, an integer m with m/N bounded away from 0 and 1, a field K , nonzero weights $\rho(t) \in K^\times$, and columns*

$$v_t = \rho(t)(1, t, \dots, t^{R-1}) \in K^R, \quad t \in T.$$

The active support family is a first-match residual slice

$$\Omega^\circ \subseteq \{x \in \{0, 1\}^T : \sum_{t \in T} x_t = m\},$$

and the prefix map is $\Phi(x) = \sum_{t \in T} x_t v_t$. Write $\mathcal{S} = \text{im } \Phi$, $L = |\mathcal{S}|$, $M = |\Omega^\circ|$, $\bar{N} = M/L$, and $F_s = \Omega^\circ \cap \Phi^{-1}(s)$. The leaf is at the frontier if $\log \bar{N} = o(N)$.

For $q = q_N \rightarrow \infty$, define

$$\Gamma_q^{\text{ord}} = L^{-1} \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{\bar{N}} \right)^q.$$

We call q logarithmic if $q_N \leq (\log N)^A$ for some fixed A ; since every image fiber is nonempty, $L \leq M \leq 2^N$, so $\log L = o(Nq)$ for every such q .

Lemma 2.2 (Moment-max equivalence). *If $q \rightarrow \infty$ and $\log L = o(Nq)$, then*

$$L^{-1} \left(\max_s \frac{|F_s|}{\bar{N}} \right)^q \leq \Gamma_q^{\text{ord}} \leq \left(\max_s \frac{|F_s|}{\bar{N}} \right)^q.$$

Hence $\Gamma_q^{\text{ord}} \leq \exp(o(Nq))$ is equivalent to $\max_s |F_s| \leq \exp(o(N))\bar{N}$.

Proof. The upper bound replaces every summand by the maximum; the lower bound keeps a maximum summand. Taking q -th roots and using $\log L = o(Nq)$ gives the final assertion. $\square$

3. SIDON CUT

For a finite F in an abelian group, let

$$E(F) = \#\{(a, b, c, d) \in F^4 : a - b = c - d\}, \quad \Delta(F) = E(F)/|F|^3.$$

Thus $\Delta(F)$ is the probability that $a - b + c \in F$ for independent uniform $a, b, c \in F$. Whenever supports are used as an additive set, we identify $S \subseteq D$ with its incidence vector $\mathbf{1}_S \in \mathbb{Z}^D$; all Boolean energies therefore lie in this torsion-free ambient group.

For $\sigma > 0$, define the Sidon-heavy moment of a primitive leaf by

$$\Gamma_{q,\sigma}^{\text{sid}} = L^{-1} \sum_{\Delta(F_s) \leq e^{-\sigma N}} \left(\frac{|F_s|}{N} \right)^q.$$

Definition 3.1 (Paid Fourier/Sidon cell). *The Fourier/Sidon cell is paid if, for every fixed $\sigma > 0$ and every logarithmic q , one has $\Gamma_{q,\sigma}^{\text{sid}} \leq \exp(o(Nq))$. Equivalently, no positive-rate heavy fiber with exponentially small additive closure remains in the primitive residual.*

In practice this cell is paid by Fourier flatness after major arcs have been routed to algebraic cells. If $\mu = \Phi_* \text{Unif}(\Omega^\circ)$ and $\sum_{\chi \neq 1} |\hat{\mu}(\chi)| \leq \exp(o(N))$, Fourier inversion gives $L\mu(s) \leq \exp(o(N))$ for all s , hence primitive $\mathbf{Q}$. Verification of a closed ledger must separately prove that every large Fourier coefficient is a quotient, descent, dihedral, ramification, or effective-image-collapse major arc; once verified, first match assigns it to the earlier cells in Definition 1.3.

Proposition 3.2 (Energy extraction after the Sidon cut). *Assume the Fourier/Sidon cell is paid. If $\Gamma_q^{\text{ord}} \geq e^{\eta N q}$ for some fixed $\eta > 0$, then some level $s = s_N$ satisfies $|F_s| \geq e^{\eta N - o(N)} N$ and $E(F_s) \geq |F_s|^3 e^{-o(N)}$.*

Proof. For a fixed $\sigma > 0$, split Γ_q^{ord} into terms with $\Delta(F_s) \leq e^{-\sigma N}$ and the complementary terms. The first part is $\Gamma_{q,\sigma}^{\text{sid}} = \exp(o(Nq))$, so a complementary term has $(|F_s|/N)^q \geq e^{\eta N q - o(Nq)}$. Hence $|F_s| \geq e^{\eta N - o(N)} N$ and $\Delta(F_s) > e^{-\sigma N}$.

For completeness, apply this assertion successively with $\sigma_j = 1/j$. Along the subsequence on which the moment excess holds, choose the j -th index after the threshold at which the paid estimate for σ_j is valid. The resulting diagonal subsequence has $\sigma_j \downarrow 0$, and therefore $E(F_s) \geq |F_s|^3 e^{-o(N)}$. $\square$

4. BOOLEAN ADDITIVE COMBINATORICS

We use the energy form of Balog–Szemerédi–Gowers [BS94, Gow98]. Its precise polynomial exponents are irrelevant here.

Theorem 4.1 (Balog–Szemerédi–Gowers). *If A is a finite subset of an abelian group and $E(A) \geq |A|^3/K$, then some $A' \subseteq A$ satisfies $|A'| \geq K^{-C}|A|$ and $|A' - A'| \leq K^C|A'|$, where C is absolute.*

We also use the quasicube sumset theorem of Matolcsi–Ruzsa–Shakan–Zhelezov [MRSZ20] in the sharpened form of Green–Matolcsi–Ruzsa–Shakan–Zhelezov [GMRSZ20]: if $U \subseteq \{0, 1\}^d$, then $|P + Q + U| \geq |P|^{1/2}|Q|^{1/2}|U|$ for finite nonempty $P, Q \subseteq \mathbb{Z}^d$.

Theorem 4.2 (Quasicube difference growth). *Every finite $A \subseteq \{0, 1\}^N$ satisfies $|A - A| \geq |A|^{3/2}$.*

Proof. Apply the quasicube theorem with $U = A$, $P = -A$, and $Q = \{0\}$. $\square$

Proposition 4.3 (No large high-energy Boolean fiber). *There is no sequence $F_N \subseteq \{0, 1\}^N$ with $|F_N| \geq e^{cN - o(N)}$ and $E(F_N) \geq |F_N|^3 e^{-o(N)}$ for fixed $c > 0$.*

Proof. By Theorem 4.1 with $K = e^{o(N)}$, a subset $F'_N \subseteq F_N$ has $|F'_N| \geq e^{cN - o(N)}$ and $|F'_N - F'_N| \leq e^{o(N)}|F'_N|$. But Theorem 4.2 gives $|F'_N - F'_N| \geq |F'_N|^{3/2}$, so $|F'_N|^{1/2} \leq e^{o(N)}$, a contradiction. $\square$

Corollary 4.4 (Quantitative Boolean energy bound). *There is an absolute constant $A > 0$ such that, for every $F \subseteq \{0, 1\}^N$ and $K \geq 1$,*

$$E(F) \geq |F|^3/K \implies |F| \leq K^A.$$

Proof. Retain the polynomial dependence in Theorem 4.1. It gives $F' \subseteq F$ with $|F'| \geq K^{-C}|F|$ and $|F' - F| \leq K^C|F'|$. By Theorem 4.2, $|F'|^{3/2} \leq K^C|F'|$, hence $|F'| \leq K^{2C}$ and $|F| \leq K^{3C}$. Take $A = 3C$. $\square$

Theorem 4.5 (Primitive Q after the Sidon cut). *In every frontier primitive leaf with paid Fourier/Sidon cell, $\max_s |F_s| \leq \exp(o(N))\bar{N}$.*

Proof. If not, Lemma 2.2 applies because $L \leq M \leq 2^N$, and gives $\Gamma_q^{\text{ord}} \geq e^{\eta N q}$ for some fixed $\eta > 0$ along a subsequence and some logarithmic q . Proposition 3.2 then gives a fiber $F_s \subseteq \{0, 1\}^T$ with $|F_s| \geq e^{\eta N - o(N)}\bar{N} = e^{cN - o(N)}$ and $E(F_s) \geq |F_s|^3 e^{-o(N)}$, because $\log \bar{N} = o(N)$. This contradicts Proposition 4.3. $\square$

5. A POLYNOMIAL-FIELD OBSTRUCTION

For $S \in \binom{D}{a}$, write

$$Q_S(X) = \prod_{t \in S} (X - t) = X^a + q_1(S)X^{a-1} + \cdots + q_a(S), \quad \Phi_w(S) = (q_1(S), \dots, q_w(S)),$$

and put $\mathcal{F}_z = \Phi_w^{-1}(z)$ and $\bar{N}_1 = \binom{n}{a} |\mathbb{B}|^{-w}$. To distinguish the unrestricted identity prefix from a routed primitive leaf, define

$$(5.1) \quad \widetilde{\Gamma}_{r,\sigma}^{\text{sid}} = |\mathbb{B}|^{-w} \sum_{\substack{z \in \mathbb{B}^w: \mathcal{F}_z \neq \emptyset \\ \Delta(\mathcal{F}_z) \leq e^{-\sigma n}}} \left(\frac{|\mathcal{F}_z|}{\bar{N}_1} \right)^r.$$

Theorem 5.1 (Smooth quotient, Sidon, and MCA obstruction). *Fix $0 < \alpha < 1/2$. Let $p \rightarrow \infty$ through odd primes, put*

$$n = 2(p-1), \quad \mathbb{B}_0 = \mathbb{F}_p, \quad \mathbb{B} = \mathbb{F}_{p^2},$$

let $H \leq \mathbb{B}^\times$ have order n , choose a generator θ of $\mathbb{B}^\times$, and set $D = \theta H$. Choose even $a = a_p$ with $a/n \rightarrow \alpha$, let w be the largest even integer not exceeding $\log_{|\mathbb{B}|} \binom{n}{a}$, and put $k = a - w - 1$. Then

$$(5.2) \quad \log |\mathbb{B}| = O(\log n) = o(n), \quad p > a, \quad w = \Theta(n/\log n), \quad 1 \leq \bar{N}_1 < |\mathbb{B}|^2 = e^{o(n)}.$$

For some $z \in \mathbb{B}^w$,

$$(5.3) \quad |\mathcal{F}_z| \geq \binom{n/2}{a/2} p^{-w/2} = \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right), \quad h(x) = -x \log x - (1-x) \log(1-x).$$

The quantity before pigeonholing is the natural average

$$\bar{N}_{\text{sq}} = \binom{n/2}{a/2} |\mathbb{B}_0|^{-w/2}$$

of the complete-square quotient profile. Thus every exhaustive profile atlas containing this quotient has $\mathfrak{E}_n(a) \geq \bar{N}_{\text{sq}} = \exp((h(\alpha)/4 + o(1))n)$, whereas $\bar{N}_1 = e^{o(n)}$; in particular the row is not identity-dominant there. There is $\sigma = \sigma(\alpha) > 0$ such that $\Delta(\mathcal{F}_z) \leq e^{-\sigma n}$, and every logarithmic $r = r_n \rightarrow \infty$ satisfies

$$(5.4) \quad \liminf_{n \rightarrow \infty} \frac{1}{nr} \log \widetilde{\Gamma}_{r,\sigma}^{\text{sid}} \geq \frac{1}{4}h(\alpha) > 0.$$

Finally, every finite extension $\mathbb{F}/\mathbb{B}$ with

$$(5.5) \quad |\mathbb{F}| - n > k \binom{|\mathcal{F}_z|}{2}$$

satisfies

$$(5.6) \quad B_{\text{RS}_{\mathbb{F}}(D,k)}^{\text{MCA}}(a) \geq |\mathcal{F}_z| \geq \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right).$$

Such extensions exist with degree $O(n/\log n)$. Hence smoothness and $\log |\mathbb{B}| = o(n)$ do not imply an identity-normalized Sidon payment or an identity-scale MCA numerator bound. Indeed,

$$B_{\text{RS}_{\mathbb{F}}(D,k)}^{\text{MCA}}(a) = \exp(\Omega(n)) \quad \text{while} \quad \bar{N}_1 = \exp(o(n)).$$

Proof. The cyclic group $\mathbb{B}^\times$ contains H , and H^2 has order $p-1$; hence $H^2 = \mathbb{B}_0^\times$. Thus $D^2 = \theta^2 \mathbb{B}_0^\times$, and squaring is two-to-one on D . The roots of $1 + X^2$ have order four and lie in H , whereas the proper coset $D = \theta H$ is disjoint from H . Since $a/n \rightarrow \alpha < 1/2$, eventually $a < p$. Stirling's formula and the even rounding give $w = nh(\alpha)/(2 \log p) + o(n/\log n)$ and $0 \leq \log_{|\mathbb{B}|} \binom{n}{a} - w < 2$, proving (5.2).

For $E \in \binom{D^2}{a/2}$, let $S_E = \{x \in D : x^2 \in E\}$. Its locator is

$$Q_{S_E}(X) = \prod_{y \in E} (X^2 - y) = X^a - e_1(E)X^{a-2} + e_2(E)X^{a-4} - \dots$$

The odd-gap coefficients vanish, while the coefficient in gap $2j$ lies in $\theta^{2j} \mathbb{B}_0$. Hence these supports occupy at most $p^{w/2}$ prefixes. Pigeonholing gives the first inequality in (5.3); moreover $\log \binom{n/2}{a/2} = \frac{1}{2}nh(\alpha) + o(n)$ and $(w/2) \log p = \frac{1}{4}nh(\alpha) + o(n)$, proving the exponent.

Let A be the constant in Corollary 4.4, and choose $0 < \sigma < h(\alpha)/(8A)$. If $\Delta(\mathcal{F}_z) > e^{-\sigma n}$, that corollary with $K = e^{\sigma n}$ gives $|\mathcal{F}_z| \leq e^{A\sigma n}$, contradicting (5.3). Keeping this single summand in (5.1) gives

$$\log \widetilde{\Gamma}_{r,\sigma}^{\text{sid}} \geq -w \log |\mathbb{B}| + r(\log |\mathcal{F}_z| - \log \bar{N}_1) = -nh(\alpha) + o(n) + r\left(\frac{1}{4}h(\alpha)n + o(n)\right),$$

which proves (5.4). Because $w < a < p$, Newton identities give an invertible triangular change between these locator coefficients and the first w power sums.

Set $U_z = X^a + z_1 X^{a-1} + \dots + z_w X^{a-w}$ and $P_S = U_z - Q_S$. Then $\deg P_S \leq k$, and for distinct $S, T \in \mathcal{F}_z$, the nonzero polynomial $Q_S - Q_T$ has degree at most k . Condition (5.5) leaves $\lambda \in \mathbb{F} \setminus D$ at which all $Q_S(\lambda)$ are distinct. On D put $r_1(x) = U_z(x)/(x - \lambda)$ and $r_2(x) = -1/(x - \lambda)$. For $\gamma_S = P_S(\lambda)$ and $x \in S$,

$$r_1(x) + \gamma_S r_2(x) = \frac{P_S(x) - P_S(\lambda)}{x - \lambda},$$

which is a polynomial of degree less than k . The slopes are distinct. Moreover r_2 cannot agree with a degree-less-than- k polynomial on $k+1$ points, since then $(X - \lambda)q(X) + 1$ would have $k+1$ roots, degree at most k , and value one at λ . Thus all γ_S are MCA-bad. Finally $\log |\mathcal{F}_z| = O(n)$ and $\log |\mathbb{B}| = \Theta(\log n)$, so an extension of degree $O(n/\log n)$ suffices. $\square$

Corollary 5.2 (Circle obstruction). *In Theorem 5.1, put $u = (k-1)/2 = (a-w-2)/2$ and take the inverse stereographic image of D on $x^2 + y^2 = 1$. Let $C_{\text{circ}}(u)$ be the evaluation, over $\mathbb{F}$, of*

$$\{f_0(x) + yf_1(x) : \deg f_0 \leq u, \deg f_1 \leq u-1\}$$

on that circle set. Then $C_{\text{circ}}(u)$ is diagonally equivalent to $\text{RS}_{\mathbb{F}}(D, k)$; hence it has the same MCA-bad slopes and the same lower bound.

Proof. The inverse parametrization is $x = (1 - t^2)/(1 + t^2)$, $y = 2t/(1 + t^2)$. Its denominator is nonzero on D by the order-four argument above. Modulo $x^2 + y^2 - 1$, the degree- u space is $\{f_0(x) + yf_1(x) : \deg f_0 \leq u, \deg f_1 \leq u-1\}$. After substitution and multiplication by $(1 + t^2)^u$, it becomes exactly the polynomials in t of degree at most $2u$: substitution is injective in the localized coordinate ring, and both spaces have dimension $2u+1$. This coordinatewise nonzero multiplier preserves all support equalities and slopes. Finally, a, w are even, so $k = a - w - 1 = 2u + 1$. $\square$

Remark 5.3 (Stabilizer deletion is insufficient). Fix $x_0 \in D$ and use $S_E = \{x_0\} \cup \{x \in D : x^2 \in E\}$, where $E \in \binom{D^2 \setminus \{x_0^2\}}{a/2}$. At support size $a + 1$ and depth w , some prefix z has a subfamily

$$\mathcal{F}_z^{\text{ap}} \subseteq \Phi_w^{-1}(z), \quad |\mathcal{F}_z^{\text{ap}}| \geq \binom{n/2 - 1}{a/2} p^{-w/2} = \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right).$$

Here every member has trivial multiplicative stabilizer. Indeed a stabilizer of a nonempty subset of D lies in $DD^{-1} = H$, and each S_E has the unique incomplete orbit $\{x_0, -x_0\}$ under -1 . Its stabilizer must preserve this pair and is therefore ± 1 , while -1 does not preserve S_E . Moreover $\binom{n}{a+1} |\mathbb{B}|^{-w} = e^{o(n)}$, so Corollary 4.4 gives $\Delta(\mathcal{F}_z^{\text{ap}}) \leq e^{-\sigma n}$ for some fixed $\sigma > 0$. This is a Sidon-heavy fiber inside the trivial- H -stabilizer residual. It is a planted remainder profile, so aperiodicity alone does not make the unrestricted identity ledger valid.

6. MCA COMPILER

Lemma 6.1 (Primitive add-back). *For a closed-ledger row, let $N_\lambda(s)$ be the residual prefix fibers in the primitive profile λ , after all algebraic cells have been removed. Then*

$$\sum_{\lambda \text{ primitive}} \max_s N_\lambda(s) \leq \exp(o(n)) \mathfrak{E}_n(a_n).$$

Proof. Primitive leaves satisfy Theorem 4.5, so their maximal fibers are $\exp(o(n))M_\lambda/L_\lambda$. The residual-to-full comparison in Definition 1.3 bounds this by $\exp(o(n))\bar{N}_\lambda$, which occurs in (1.1). Summing over the subexponential first-match profile family preserves the same scale. $\square$

Lemma 6.2 (Q pays shift pairs). *If $\sum_s N(s) = M$, $\bar{N} = M/L$, and $\max_s N(s) \leq \kappa \bar{N}$, then $M^{-1} \sum_s N(s)^2 \leq \kappa \bar{N}$.*

Proof. $\sum_s N(s)^2 \leq (\max_s N(s)) \sum_s N(s) \leq \kappa \bar{N} M$. $\square$

Remark 6.3 (Q and SP do not count rays). The preceding lemma is a support-pair identity, not a ray compiler. Keeping the same fibers $N(s)$, an abstract incidence can assign distinct slopes to selected supports or assign one slope to all of them without changing $\max_s N(s)$ or $\sum_s N(s)^2$. An RS numerator bound therefore requires the additional algebraic projection estimate RC.

Theorem 6.4 (Closed-ledger upper bound). *If a smooth RS row sequence is closed-ledger and satisfies RC at a_n , then*

$$B_{C_n}^{\text{MCA}}(a_n) \leq \exp(o(n)) \mathfrak{E}_n(a_n).$$

Proof. The nonprimitive cells are paid by definition. The primitive prefix leaves obey Q by Theorem 4.5 and add back by Lemma 6.1; their support-pair second moment is bounded by Lemma 6.2. By Remark 6.3, this does not itself bound slopes. RC supplies the required slope-image estimate, and first-match disjointization sums the disjoint slope classes. $\square$

7. LOWER SIDE AND FRONTIER

Proposition 7.1 (Collision-aware identity lower bound). *Let $D \subseteq \mathbb{B} \subseteq \mathbb{F}$, let $C = \text{RS}_{\mathbb{F}}(D, k)$, $q = |\mathbb{F}| > n$, and $k + 1 \leq m \leq n$. Put*

$$w = m - k - 1, \quad L = \left\lceil \binom{n}{m} |\mathbb{B}|^{-w} \right\rceil.$$

Then

$$(7.1) \quad B_C^{\text{MCA}}(m) \geq \left\lceil \frac{L(q - n)}{q - n + k(L - 1)} \right\rceil.$$

Proof. Pigeonhole the first w locator coefficients to obtain L supports with common prefix z . If $Q_S = \prod_{x \in S} (X - x)$ and $U_z = X^m + z_1 X^{m-1} + \dots + z_w X^{m-w}$, then $P_S = U_z - Q_S$ has degree at most k . At a pole $\lambda \in \mathbb{F} \setminus D$, the simple-pole line used in the proof of Theorem 5.1 makes every distinct value $P_S(\lambda)$ an MCA-bad slope.

For $S \neq T$, equality $P_S(\lambda) = P_T(\lambda)$ occurs at at most k poles. Averaging over the $q - n$ poles gives one pole with at most $k \binom{L}{2} / (q - n)$ colliding pairs. If its value multiplicities are $m_1, \dots, m_M$, then

$$\sum_i m_i^2 \leq L + \frac{kL(L-1)}{q-n}.$$

Cauchy–Schwarz, $L^2 \leq M \sum_i m_i^2$, proves (7.1). This is the quantitative form of the prefix-pole construction in [Cho26a, Cho26b]. $\square$

Proof of Theorem 1.7. Condition (i) and Theorem 6.4 make $a_{+,n}$ safe, while condition (ii) makes $a_{-,n}$ unsafe. Since $B_{C_n}^{\text{MCA}}(a)$ is nonincreasing in a , the first safe agreement satisfies $a_{-,n} < a_n^* \leq a_{+,n}$, proving (1.2). If the endpoints are $o(n)$ apart around $k_n + 1 + g_n n$, division by n gives $\delta_n^* = 1 - \rho_n - g_n + o(1)$. The displayed identity choice of $U_n(a)$ is Proposition 7.1.

For $a_n = k_n + 1 + g_n n + O(1)$, Stirling’s formula gives

$$\log_2 \bar{N}_{n,a_n} = nF_n(g) + o(n).$$

Thus $g_{T,n}$ is the exponential-scale right boundary at which the identity average has target rate τ_n . The separate endpoint hypothesis converts this exponential comparison into the exact safe and unsafe inequalities, accounting for the tangent term, compiler loss, integer rounding, and pole collisions. The preceding bracket gives the target-adjusted formula. Finally, if $\tau_n = o(1)$, concavity of F_n , together with $F_n(0) = H_2(\rho_n)$ bounded away from zero, shows that its right τ_n -level boundary differs by $o(1)$ from its right zero-level boundary $g^*(\rho_n, \beta_n)$. This proves (1.4). $\square$

Remark 7.2 (Scope of the main statement). Theorem 5.1 shows that smooth/circle geometry and $\log |\mathbb{B}_n| = o(n)$ do not imply the identity-scale upper bound: a degree-two quotient over a proper subfield already has an exponentially larger natural prefix budget. Thus the unconditional conclusions here are the obstruction and the high-energy elimination. The frontier formula is the identity-dominant specialization of the profile-envelope compiler. For unrestricted rows, the remaining requirements are exhaustive quotient/remainder routing with residual-to-full profile comparison, a Sidon payment on the resulting primitive residual, and a higher-dimensional transverse-ray compiler. Because the constructed depth satisfies $w/n \rightarrow 0$, the example alone does not disprove the coarse $o(1)$ -radius formula for subexponential targets; it disproves the identity-scale numerator bound and hence any unconditional derivation of that formula from the identity prefix alone.

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